Sara Ghasvarianjahromi, PhD

ML Research Engineer | LLMs, Information Retrieval & Robust Machine Learning

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Summary

Ph.D. in Electrical Engineering specializing in machine learning, semantic retrieval, and robust decentralized computation. Research experience in retrieval under uncertainty, Byzantine-resilient computation, and machine learning–based optimization, with additional applied work in wireless network optimization. Proficient in Python, PyTorch, mathematical optimization, and statistical modeling, with multiple peer-reviewed IEEE publications.

Skills

Machine Learning: Large Language Models (LLMs), Semantic Retrieval, Retrieval-Augmented Generation (RAG), Decentralized Learning, Adversarial Robustness, Deep Learning, AI Agents

Programming: Python, MATLAB

ML Frameworks & Libraries: PyTorch, Hugging Face Transformers, pandas, NumPy, scikit-learn

Optimization & Modeling: Gurobi, Convex Optimization, Statistical Modeling, Probabilistic Modeling

Tools: Git, Linux, VS Code, LaTeX

Experience

New Jersey Institute of Technology, NJ – **Research Assistant** Jan 2022 – May 2026

- Built a context-aware semantic retrieval system using pretrained embeddings to improve retrieval accuracy under noisy and lossy communication conditions.
- Developed and validated decentralized ML architectures with adversarial defenses in PyTorch, achieving 30–50% higher accuracy than state-of-the-art baselines and up to 25% improvement under adversarial noise on MNIST and CIFAR-10.
- Implemented a fully decentralized sparse matrix multiplication framework with formal robustness guarantees, enabling scalable coordination-free distributed computation.
- Derived probabilistic performance bounds to characterize system behavior under uncertainty.

Futurewei Technologies, NJ – **ML Research Intern** May 2025 – Aug 2025

- Built an end-to-end AI/ML pipeline for QoS optimization using 3GPP network analytics, reducing energy consumption by 18–25% over fixed-latency baselines and 40–50% over non-energy-aware systems while reducing latency by 30–40%.
- Trained neural network models in PyTorch to infer adaptive, per-user QoS recommendations from network telemetry, replacing rule-based decision logic with data-driven inference.
- Designed a Gurobi-based optimization pipeline for energy-aware path selection and transmit-power control, jointly minimizing latency and energy under strict QoS constraints.

Ozyegin University, Istanbul, Turkey – **Research Assistant** Jan 2018 – Jul 2021

- Developed convex optimization algorithms for energy-efficient wireless systems, extending device battery lifetime by up to 110%.

Projects

AI Job Scout Agent

- Built a multi-agent AI job search assistant integrating LLM reasoning, semantic ranking, SQLite persistence, Git-based development, and automated cover letter/LinkedIn message generation.
- Developed in Python using the Gemini API, modular agent architecture, and multiple job data sources.

Parameter-Efficient LLM Fine-Tuning

- Fine-tuned DistilBERT using LoRA (PEFT) on SST-2 sentiment classification with Hugging Face Transformers and PyTorch.

Education

PhD in Electrical Engineering, New Jersey Institute of Technology Newark, NJ
Jan 2022 – May 2026

Dissertation: Reliable Machine Learning and Information Retrieval under Uncertainty

Certifications & Professional Development

Google AI Agents Workshop – Built and deployed LLM-based agentic AI applications using Google's Agent Development Kit (ADK) and Vertex AI during an intensive hands-on workshop. (Google, Jun 2026)

Google Machine Learning Crash Course (badges) – LLMs, embeddings, and production ML systems modules (Google, 2025–2026)

Selected Publications

Full list on **Google Scholar**.

1. **S. Ghasvarianjahromi**, J. Barr, Y. Yakimenka, J. Kliewer, "Context-Aware Search and Retrieval Under Token Erasure," Submitted, 2026.
2. **S. Ghasvarianjahromi**, A. Kiani, A. Xiang, J. Kaippallimalil, T. Saboorian, N. Ansari, "AI/ML-based QoS Recommendations for Energy Optimization in 6G Networks," ***IEEE Netw. Lett.***, 2025.
3. M. Bakshi, **S. Ghasvarianjahromi**, Y. Yakimenka, A. Beemer, O. Kosut, J. Kliewer, "VALID: A Validated Algorithm for Learning in Decentralized Networks with Possible Adversarial Presence," ***IEEE Intern. Symp. Info. Theo. (ISIT)***, 2024.

Professional Service

IEEE Reviewer – Wireless Communications Letters, Communications Letters, Transactions on Communications, Transactions on Vehicular Technology, Transactions on Wireless Communications, Elsevier Physical Communications